

Question 1:

If the chord joining the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

- (A) 292
- (B) 288
- (C) 284
- (D) 280

Question 2:

Let $P(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance between the lines $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$ is equal to

- (A) $7\sqrt{5}/4$
- (B) $4\sqrt{5}/7$
- (C) $2\sqrt{7}/4$
- (D) $4\sqrt{7}/5$

Question 3:

If $A = \begin{bmatrix} 2 & 3 & 5 \end{bmatrix}$, then the determinant of the matrix $A^{2025} - 3A^{2024} + A^{2023}$ is

- (A) 28
- (B) 16
- (C) 24
- (D) 12

Question 4:

Let the relation R on the set $M = \{1, 2, 3, \dots, 16\}$ be given by $R = \{(x, y) : 4y = 5x - 3, x, y \in M\}$. Then the minimum number of elements required to be added in R , in order to make the relation symmetric, is equal to

- (A) 3
- (B) 4
- (C) 2
- (D) 1

Question 5:

If $\cos 248^\circ - \sin 12^\circ / \sin 24^\circ - \sin 26^\circ = \alpha + \beta \sqrt{5/2}$,
where $\alpha, \beta \in \mathbb{N}$, then the value of $\alpha + \beta$ is _____.

Question 6:

A meter bridge with two resistances R_1 and R_2 as shown in figure was balanced (null point) at 40 cm from the point P.

The null point changed to 50 cm from the point P, when a 16Ω resistance is connected in parallel to R_2 .

The values of resistances R_1 and R_2 are

- (A) $R_2 = 4\Omega, R_1 = 43\Omega$
- (B) $R_2 = 16\Omega, R_1 = 163\Omega$
- (C) $R_2 = 8\Omega, R_1 = 163\Omega$
- (D) $R_2 = 12\Omega, R_1 = 123\Omega$

Question 7:

Net gravitational force at the center of a square is found to be F_1 when four particles having masses $M, 2M, 3M$ and $4M$ are placed at the four corners of the square as shown in figure, and it is F_2 when the positions of $3M$ and $4M$ are interchanged.

The ratio $F_1 / F_2 = \alpha / \sqrt{5}$.

The value of α is

- (A) 1
- (B) 3
- (C) $2\sqrt{5}$
- (D) 2

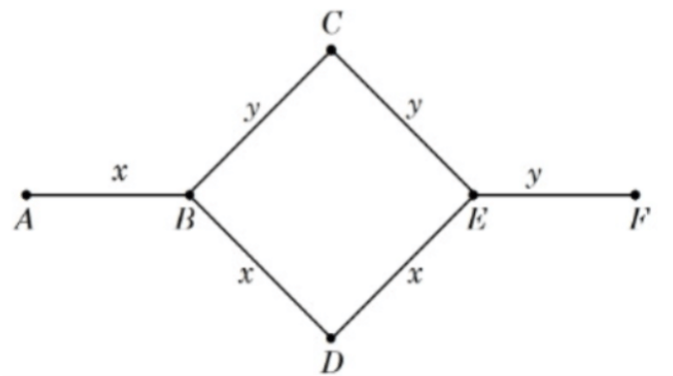
Question 8:

A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed, a gap of 1 cm is introduced between the lenses by moving the concave lens away, which leads to a change in magnification of total lens system to m_2 . The value of $m_1 m_2$ is

- (A) $25/27$
- (B) $3/2$
- (C) $5/27$
- (D) $5/9$

Question 9:

Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at 100°C and 40°C respectively. Given the thermal conductivity of rod x is three times of that of rod y , the temperature at junction points B and E are (close to):



- (A) 60°C and 45°C respectively
- (B) 89°C and 73°C respectively
- (C) 80°C and 70°C respectively
- (D) 80°C and 60°C respectively

Question 10:

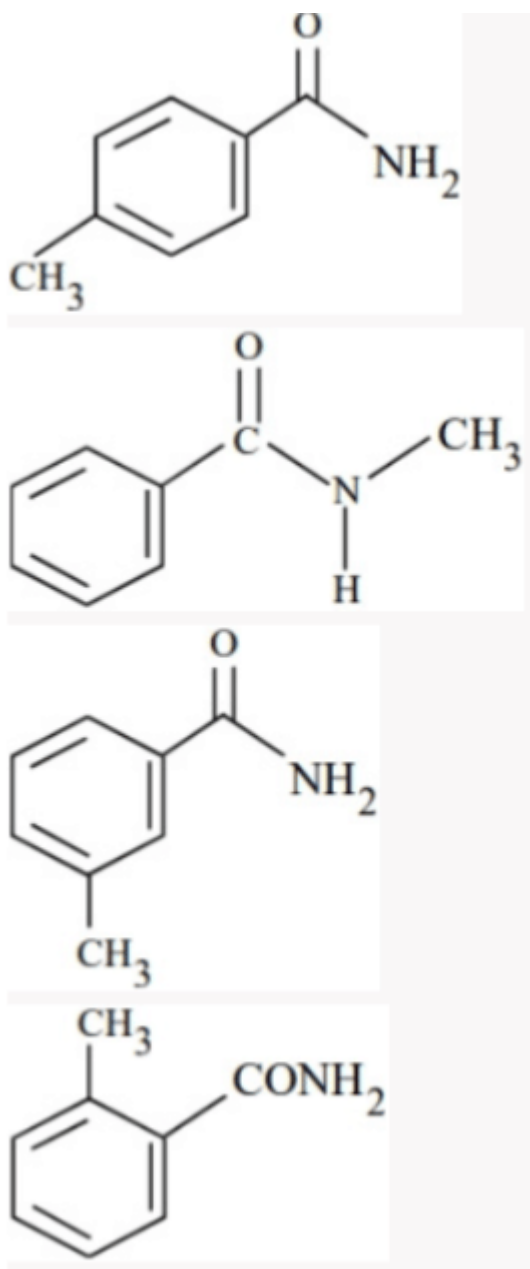
Inductance of a coil with 10^4 turns is 10mH and it is connected to a DC source of 10V with internal resistance 10Ω . The energy density in the inductor when the current reaches $(1e)$ of its maximum value is $\alpha\pi \times 10^2 \text{ Jm}^{-3}$.

The value of α is _____.

($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)

Question 11:

A is a neutral organic compound (M.F.: $\text{C}_8\text{H}_9\text{ON}$). On treatment with aqueous $\text{Br}_2/\text{HO}-$, A forms a compound B which is soluble in dilute acid. B on treatment with aqueous NaNO_2/HCl ($0-5^\circ\text{C}$) produces a compound C which on treatment with CuCN/NaCN produces D. Hydrolysis of D produces E which is also obtainable from the hydrolysis of A. E on treatment with acidified KMnO_4 produces F. F contains two different types of hydrogen atoms. The structure of A is



Question 12:

$\text{A} \rightarrow \text{Product}$ (**First order reaction**).

Three sets of experiments were performed for a reaction under similar experimental conditions:

Run1 $\Rightarrow$ 100 mL of 10 M solution of reactant A
 Run2 $\Rightarrow$ 200 mL of 10 M solution of reactant A
 Run3 $\Rightarrow$ 100 mL of 10 M solution of reactant A + 100 mL of H_2O

The correct variation of rate of reaction is

- (A) $\text{Run } 3 < \text{Run } 1 = \text{Run } 2$
- (B) $\text{Run } 1 = \text{Run } 2 = \text{Run } 3$
- (C) $\text{Run } 1 < \text{Run } 2 < \text{Run } 3$
- (D) $\text{Run } 3 < \text{Run } 1 < \text{Run } 2$

Question 13:

Match the LIST-I with LIST-II.

List-I		List-II	
Thermodynamic Process		Magnitude in kJ	
A.	Work done in reversible, isothermal expansion of 2 mol of ideal gas from 2 dm^3 to 20 dm^3 at 300 K.	I.	4
B.	Work done in irreversible isothermal expansion of 1 mol ideal gas from 1 m^3 to 3 m^3 at 300 K against a constant pressure of 3kPa.	II.	11.5
C.	Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature = 320 K and $\bar{C}_V = \frac{3}{2} R$.	III.	6
D.	Change in enthalpy at constant pressure of 1 mol ideal gas with change of temperature = 337 K and $\bar{C}_p = \frac{5}{2} R$.	IV.	7

Choose the correct answer from the options given below:

- (A) A-III, B-II, C-IV, D-I
- (B) A-II, B-I, C-III, D-IV
- (C) A-I, B-II, C-III, D-IV

- (D) A-II, B-III, C-I, D-IV

Question 14:

Given below are two statements:

Statement I: Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

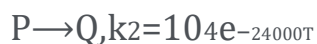
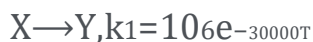
Statement II: Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose, the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Statement I is false but Statement II is true
- (B) Statement I is true but Statement II is false
- (C) Both Statement I and Statement II are false
- (D) Both Statement I and Statement II are true

Question 15:

The temperature at which the rate constants of the given below two gaseous reactions become equal is _____ K (Nearest integer).



Given: $\ln 10 = 2.303$